

growing, catalysed by initiatives like SID India's workshops and seminars.

To fully seize the moment, India needs:

- Focused investments in automotive display module clusters co-located with auto hubs (like Pune, Chennai, Manesar, Sanand).
- Co-development programmes between OEMs, EMS providers, and display tech startups.
- Accelerated adoption of automotive display testing infrastructure, including vibration, thermal cycling, humidity, and optical quality benchmarks.

A unified approach that bridges EMS expertise, automotive-grade reliability, and display innovation can propel India into the global automotive display value chain, not merely as a consumer but as a creator.

Make-in-India and PLI: Enabling or inhibiting automotive displays?

India's Make-in-India campaign and production-linked incentive (PLI) schemes have undeniably accelerated the growth of electronics and component manufacturing. Mobile phone exports have surged, and semiconductor packaging is on the rise. However, in automotive displays, the policy landscape remains fragmented and underutilised.

While displays are a critical component of both automotive and consumer electronics value chains, they have not been explicitly prioritised within most existing PLI frameworks. The India Semiconductor Mission (ISM) 1.0 policy, aimed at attracting fabrication units, has seen limited traction for display fabs to date. High capital costs, long gestation periods, and global panel oversupply have made investors cautious, particularly in the absence of committed offtake from Indian automotive OEMs or a strong local Tier-1 display ecosystem.

At the same time, the recent Electronics Component Manufacturing Scheme (ECMS) policy, though

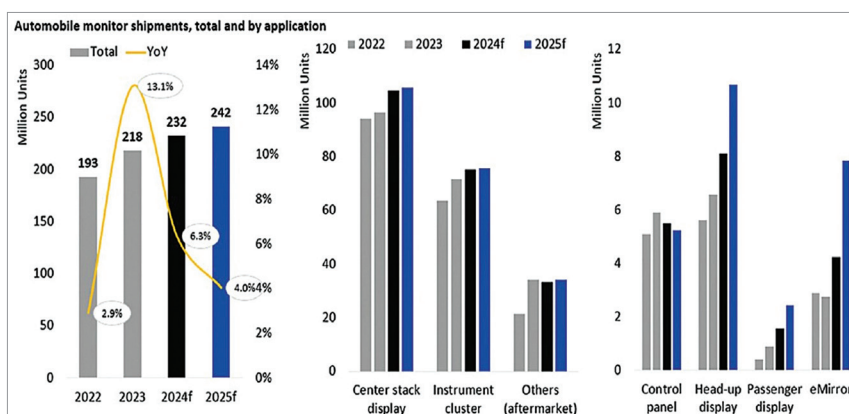


Fig. 2: The automobile monitor market

helpful, continues to suffer from inverted duty structures. Automotive display modules face significantly stricter performance requirements than consumer electronics, yet current incentives are not designed to support critical elements such as optical bonding lines, automotive-grade printed circuit board (PCB) assembly, thermal and vibration testing laboratories, or rugged display controller design—all essential parts of the value chain.

There is, therefore, a growing need to design a display-focused industrial policy that recognises:

- The strategic value of cockpit digitisation in India's mobility future
- The rising demand from domestic OEMs and Tier-1s
- The potential for mid-size EMS companies to scale up with the right policy tailwinds

A targeted automotive display sub-mission under Make-in-India could enable:

- Local module assembly co-located with auto hubs
- Incentives for validation, reliability testing, and certification labs
- Support for startups and design-led firms working on embedded HMI and UI platforms
- Integration with ECMS, SEMI-CON-PLI, and R&D grant schemes to create a full-stack ecosystem

India does not lack demand or ambition; it needs convergence and clarity in policy execution. Aligning Make-in-India incentives

with automotive display objectives could unlock a high-impact, globally relevant sector at the intersection of electronics, design, and mobility.

Academia and skills: The missing link

Behind every technological breakthrough lies a pipeline of skilled talent, and in the domain of automotive displays, this pipeline in India remains underdeveloped. The field is inherently interdisciplinary, requiring expertise in materials science, optics, electronics, embedded systems, mechanical design, and HMI. Yet, few Indian academic institutions currently offer focused coursework or laboratory exposure in display technologies, let alone automotive-grade standards.

While academic interest is increasing, curricula, infrastructure, and industry linkages remain limited.

There is a pressing need to:

- Introduce curriculum modules on display design, embedded HMI systems, and automotive UI/UX
- Create faculty development programmes in partnership with SID India and global experts
- Offer industry internships with EMS firms and automotive OEMs working on display interfaces

Organisations such as SID India have taken early steps through workshops, tutorials, and student-industry engagement. However, scaling this nationally will require a dedicated display-skills sub-mission aligned with the National Skill Development